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POCKET OTOSCOPE TO DETECT EAR INFECTIONS

A report on MINI PROJECT (22EI69P)

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BACHELOR OF ENGINEERING
IN
ELECTRONICS & INSTRUMENTATION ENGINEERING

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INTRODUCTION

A smartphone otoscope is a small, portable medical tool that helps check inside the ear using a smartphone's camera and light. Normal otoscopes are big, expensive, and not always available in poor areas. A smartphone otoscope is cheaper and easy to use because it works with phones that people already have. Using 3D modeling makes it better by helping with design, comfort, and how easy it is to build. Doctors, students, and engineers can quickly make and improve new versions.

The main idea is to attach a small lens and light to a smartphone. The phone camera takes clear pictures or videos of the ear, which can be shown on the screen, shared with doctors, or saved. 3D modeling helps make sure the attachment fits different phones and lines up perfectly with the camera.

A big advantage of using a 3D model is that you can see what the otoscope will look like before making it. You can test how parts fit, how comfortable it is to hold, and how well it works. 3D models also help make snap-fit parts that don't need glue or screws, which makes things cheaper and easier.

The 3D model is also good for learning and research. Medical students can see how it's built and how it works. Researchers can try new ideas like adding LED lights, focus controls, or apps that show live video. You can also change the design for kids by making smaller parts.

For the world, a smartphone otoscope with 3D modeling is super helpful. In far-away or poor places without many doctors, health workers can use it to take pictures and ask experts for help using the internet. Since 3D printing is cheap and phones are common, this tool can help more people stay healthy.

OBJECTIVES

We aim to develop a compact, affordable, and ergonomic otoscope with high-quality optics, focused LED illumination, a 3D-printed body, and a user-friendly web interface for real-time ear examination and image analysis.

Earlier Work

The otoscope has come a long way over the years. It all started way back in **1363**, when a French doctor named Guy de Chauliac used a simple tool called an *ear speculum* to look inside ears. But big changes started in the 1800s, when a German doctor named Anton von Troltsch made ear exams a regular part of medical checkups. Around the same time, a scientist named Hermann von Helmholtz invented the ophthalmoscope (used for eyes), which helped inspire better tools for ears too.

By the **early** 1900s, doctors added electric lights to otoscopes so they could see more clearly. Then in the 1950s, fiber optics were added, making the light even better and helping doctors spot problems more easily. Otoscopes became smaller, easier to carry, and used batteries, so doctors could take them anywhere.

Even though the basic idea stayed the same, new tech has made otoscopes much smarter and more helpful. Some of the latest upgrades include:

- **AI-Powered Otoscopes** – These use artificial intelligence to look at ear pictures and help find problems like infections or too much earwax.
- **OCT (Optical Coherence Tomography)** – A fancy new method that shows super-detailed images of the ear, even parts that normal otoscopes can't see.
- **Digital and Video Otoscopes** – These have HD cameras and can connect to phones or computers, so doctors can zoom in, save, and study images.
- **Wispr Digital Otoscope** – Made by Dr. Jim Berbee, this tool gives super clear images and is approved by the FDA.

Telemedicine Use – Now, smart otoscopes can send ear images online, so doctors can help patients from far away, even in remote areas.

METHODOLOGY

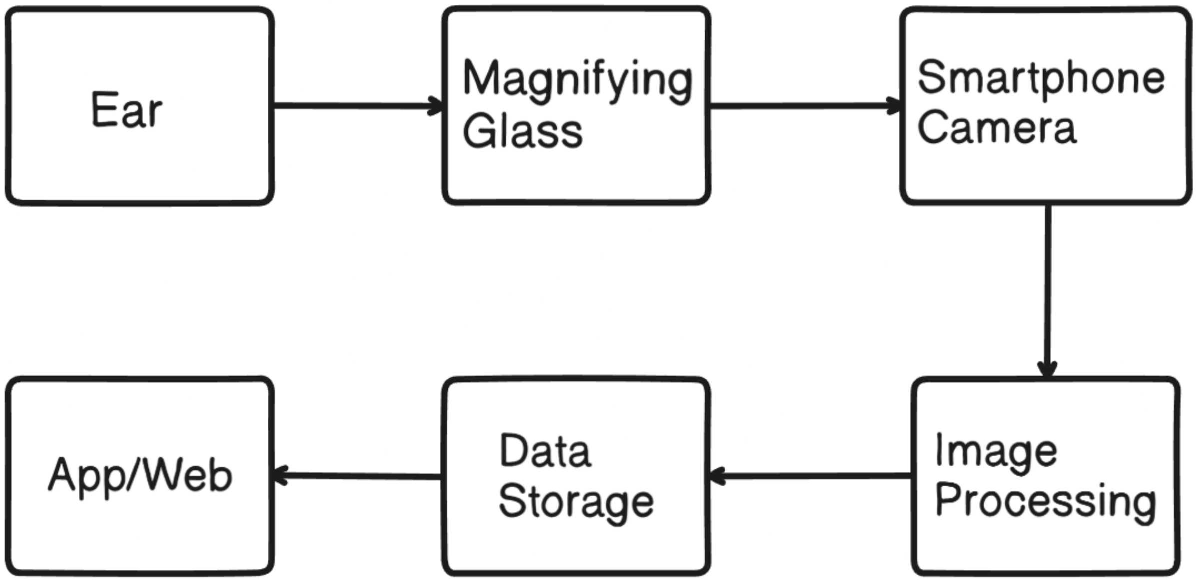

Figure 1:Block diagram

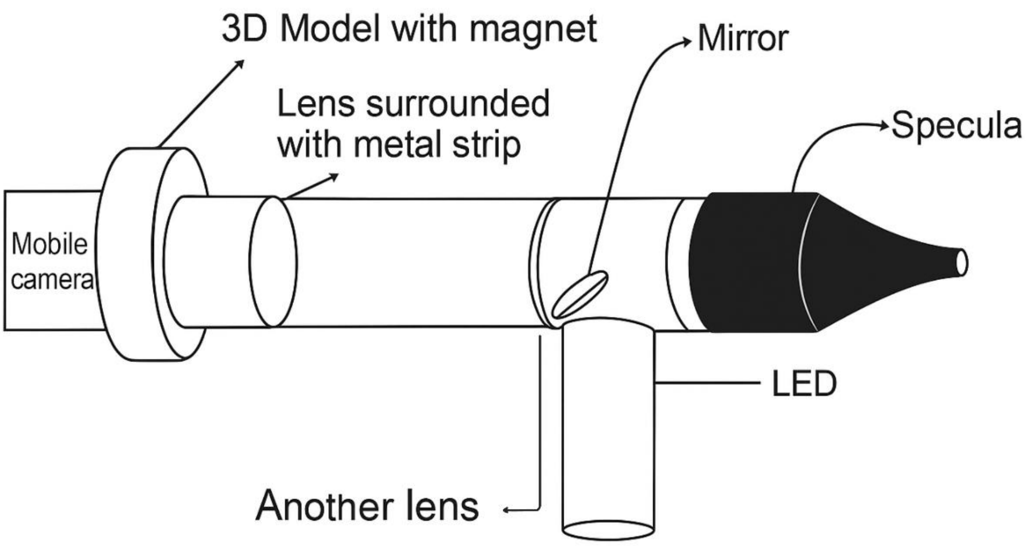

Figure 2: Model diagram

HARDWARE

Our project aims to build a custom otoscope integrated with a digital imaging system for medical use. The following hardware components have been selected and integrated based on functional requirements:

1. Lenses (Optical System)

- We are using two optical lenses to achieve the required magnification and focus.
- These lenses help in capturing a clear and zoomed-in image of the ear canal.
- The selection was based on focal length, lens diameter, and compatibility with the camera module.

2. LED Light Source

- A high-intensity white LED is fixed near the tip of the otoscope.
- It provides sufficient illumination inside the ear canal, which is essential for capturing clear images.
- The LED is powered using a portable power source and is fixed securely in the 3D-printed housing.

3. Camera Module (To be Integrated)

- A small camera module (e.g., USB or Wi-Fi endoscope camera) will be integrated into the otoscope for capturing images.
- The camera will be positioned behind the lenses for focused imaging.
- It will be connected to a PC or mobile device for real-time image display and processing.

4. 3D Printed Otoscope Body

- The body of the otoscope is custom-designed using CAD software.
- The design ensures proper alignment of the camera, lenses, and LED light.
- The model has been sent for 3D printing using durable material to provide strength and lightweight handling.

5. Power Supply & Wiring

- The LED and camera require a low-voltage DC power supply, which can be supplied via USB or battery pack.
- Internal wiring is carefully planned within the otoscope body for safety and compactness.

SOFTWARE

Overview

Earprobe is a secure, responsive, and community-driven web platform designed for ENT (Ear, Nose, Throat) specialists to capture, store, and share diagnostic ear images. The platform streamlines clinical collaboration and improves patient care by enabling easy photo capture and peer-to-peer doctor communication.

Key Features

- **Doctor Login System**
  Secure authentication to protect sensitive patient data and ensure access is limited to verified medical professionals.
- **Responsive & Animated UI**
  Built with **Shadcn components** and **Framer Motion** for a modern, mobile-first experience with smooth transitions and UI feedback.
- **Beautiful Homepage**
  A clean and engaging homepage with a minimalistic design that welcomes doctors and introduces the platform's core functionality.
- **Patient Data Form**
  Input fields to record:
  - Patient Name
  - Age
  - Phone Number
  - Ear Photo (captured directly from the camera)
- **Camera Integration**
  Direct integration with mobile device camera to capture ear images without needing third-party apps.

- **Doctor-to-Doctor Sharing**
  A community feature that allows one doctor to securely send diagnostic images and patient information to another for second opinions or consultations.

- **Data Storage and Access**
  Secure cloud-based storage for diagnostic images and patient metadata with role-based access control.

Tech Stack

- **Frontend:** React js, Shadcn UI, Tailwind CSS, Framer Motion
- **Backend:** Node.js / Express
- **Database:** MongoDB
- **Authentication:** NodeAuth
- **Image Storage:** Cloudinary

RESULT

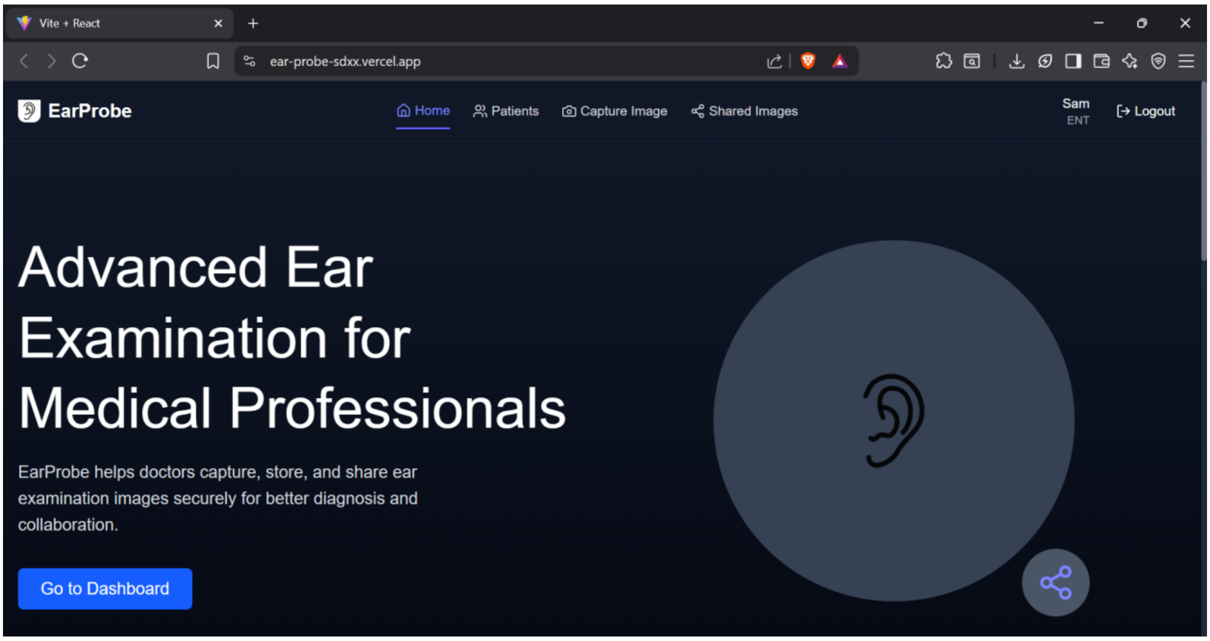

Figure 3: Home page

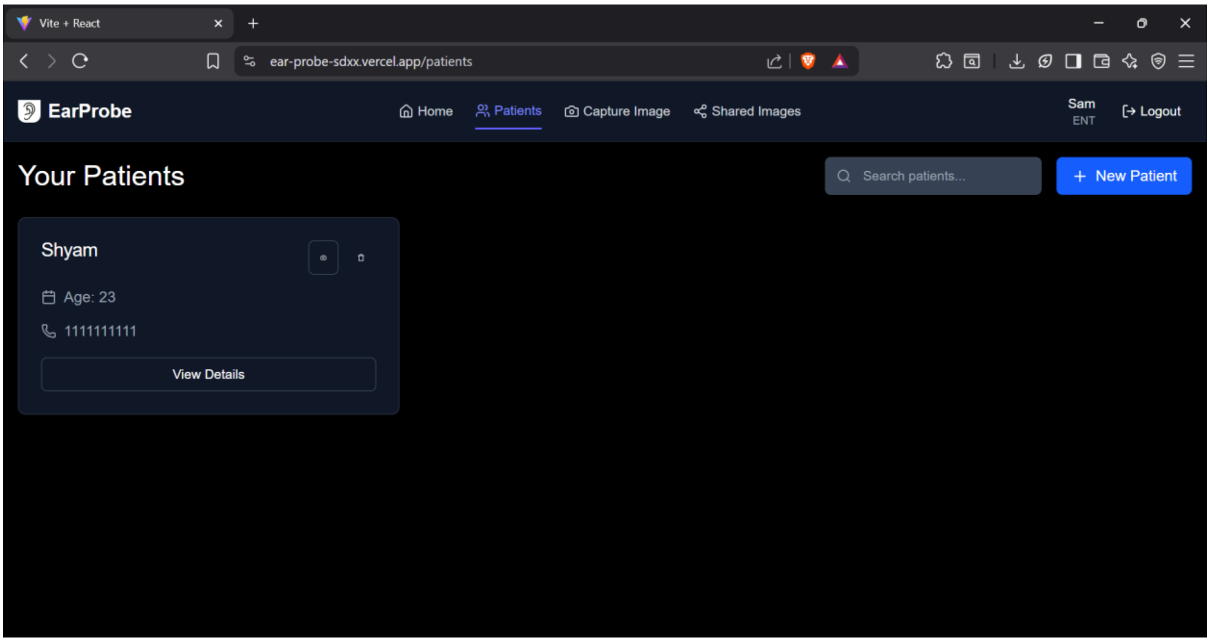

Figure 4: Patient list page

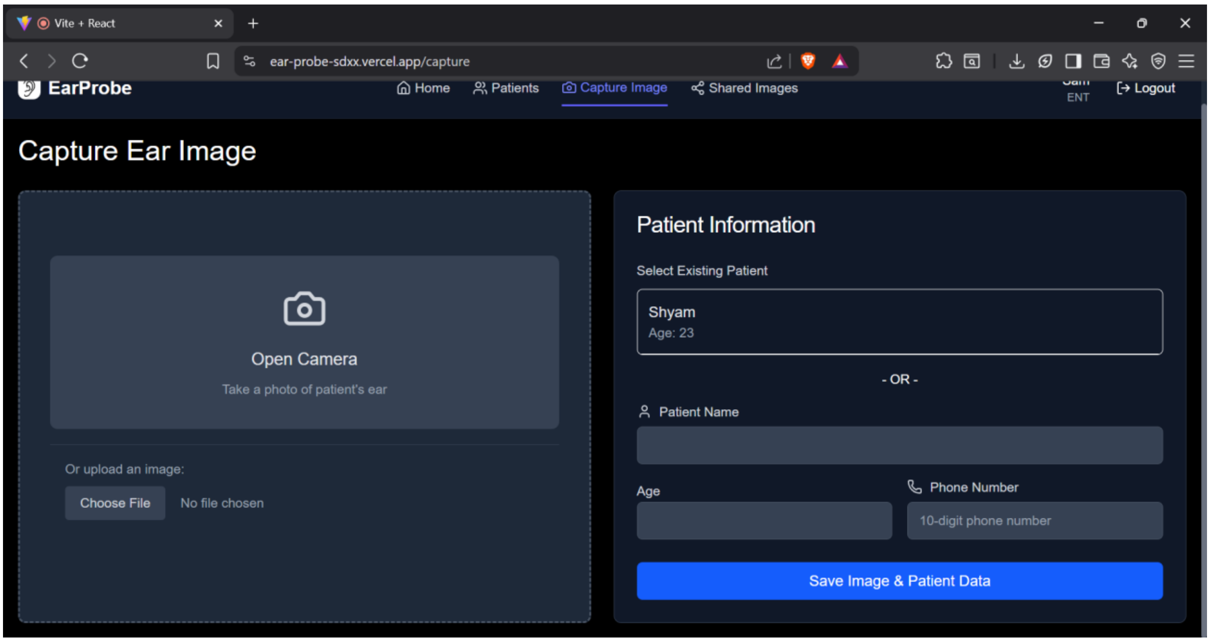

Figure 5: Capture the image page

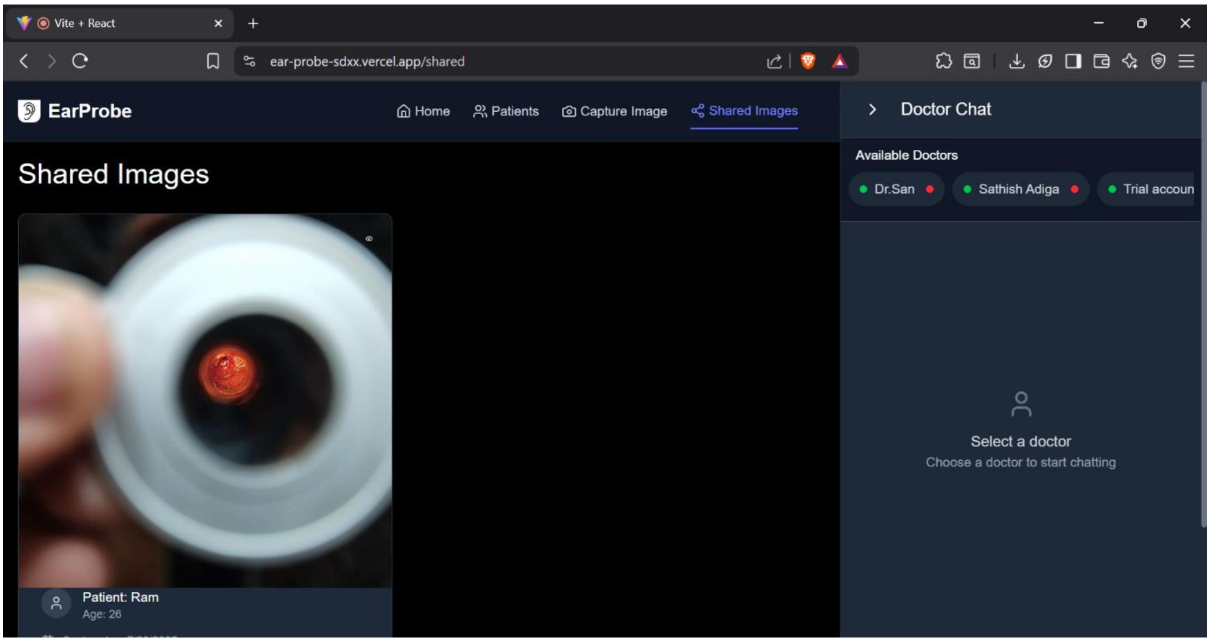

Figure 6: Sharing of the image

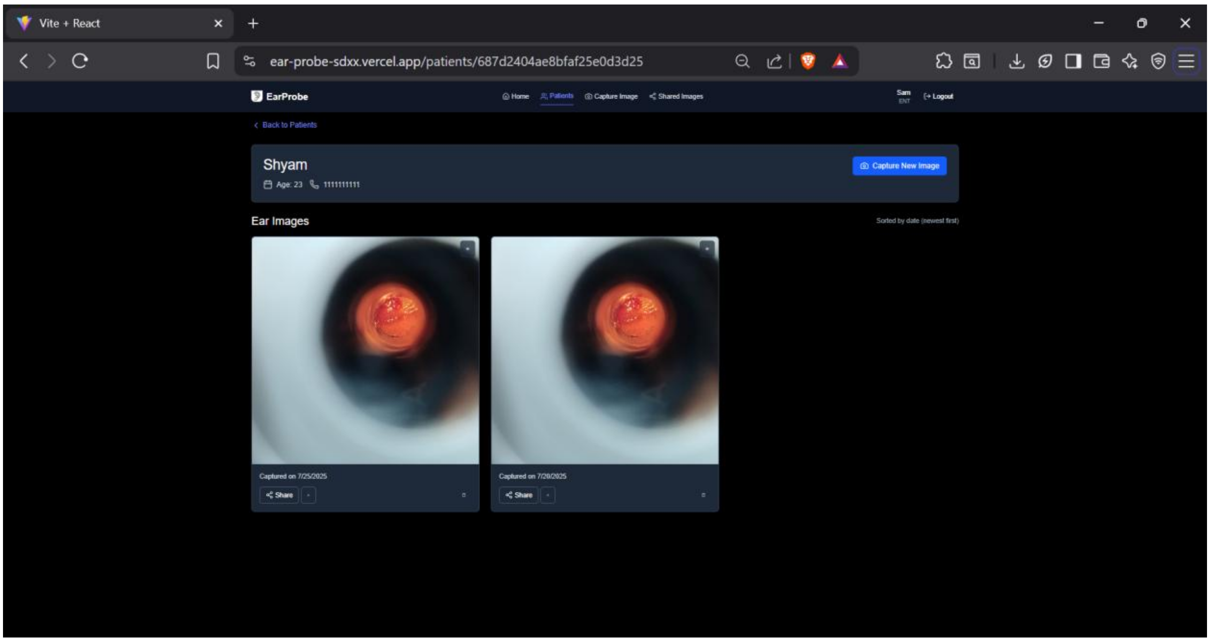

Figure 7: Comparing the old image and new image

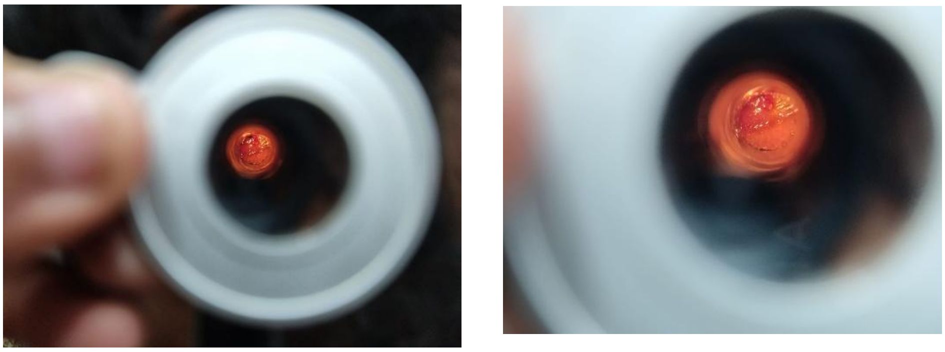

Figure 8: Captured image using our otoscope

APPLICATION

Smart otoscopes with mobile integration and variable focus are transforming ear examinations by making them more accessible, precise, and user-friendly. Here's a breakdown of their key applications:

Medical & Clinical Use

- Telemedicine: Enables remote diagnosis by transmitting live images or videos to healthcare providers.
- Patient Monitoring: Allows patients to track ear conditions over time and share updates with doctors.
- Documentation: Captures high-resolution images for medical records, teaching, or second opinions.

Home & Family Health

- Self-examination: Empowers individuals to check for earwax buildup, infections, or inflammation without visiting a clinic.
- Childcare: Parents can monitor ear health in children, especially useful for recurring ear infections.
- Pet Care: Some models are stable enough for use on pets, aiding in basic veterinary checks.

Tech & Imaging Features

- Mobile App Control: Users can adjust focus, capture images, and record videos directly from their smartphones.
- Variable Focus: Enhances clarity by adapting to different ear canal depths and obstructions like wax.
- Real-time Streaming: Offers live HD video for immediate feedback and diagnosis.

Research & Education

- Medical Training: Used in teaching environments to demonstrate ear anatomy and pathology with real-time visuals.

- Clinical Studies: Facilitates data collection for research on ear conditions and treatment outcomes.

Design & Engineering Innovations

- Attachment Devices: model features detachable speculums and universal mounts for various smartphones.
- Light Optimization: Internal reflection and lens coatings improve illumination and image quality.

Main Advantages

- Diagnosing Ear Infections: Helps detect infections in the middle and outer ear by examining redness, swelling, or fluid buildup.
- Checking for Earwax Blockage: Allows healthcare providers to assess excessive earwax that might be causing hearing issues.
- Screening for Hearing Loss: Used to inspect the ear canal and eardrum for abnormalities that could contribute to hearing impairment.
- Evaluating Eardrum Mobility: Pneumatic otoscopes can apply air pressure to check the movement of the eardrum, aiding in diagnosing middle ear conditions.
- Identifying Foreign Objects: Useful for detecting and removing objects lodged in the ear canal, especially in children.

Future Scope and Improvements

Advanced Medical Applications

- AI-Assisted Diagnosis: Some models use machine learning to detect patterns like otitis media or tympanic membrane perforations.
- 3D Ear Canal Mapping: Variable focus and depth sensing can help reconstruct 3D models for surgical planning or anatomical studies.
- Multi-Spectral Imaging: Future devices may use infrared or UV light to detect inflammation or fluid buildup not visible in standard light.

Connectivity & Data Management

- Cloud Integration: Automatically syncs patient data and images to secure cloud platforms for remote access and long-term storage.
- EHR Compatibility: Seamless integration with electronic health records for streamlined documentation and follow-up.
- Multi-Device Syncing: Enables simultaneous viewing on multiple screens—ideal for collaborative diagnosis or teaching.

Smart Features & Automation

- Auto-Focus & Stabilization: Gyroscopic sensors and adaptive optics reduce blur from hand movement, especially useful for pediatric or pet exams.
- Voice-Controlled Operation: Hands-free control via voice commands for capturing images or switching modes.
- Annotation & Analysis Tools: In-app tools allow marking areas of concern, measuring dimensions, and comparing historical images.

Engineering & Design Innovations

- Modular Attachments: Interchangeable speculums and lens modules for different use cases (ear, nose, throat).
- Universal Mounts: Clip-on or magnetic mounts compatible with a wide range of smartphones and tablets.
- Reflective Tunnel Design: Enhances illumination by channeling light through internal reflective surfaces for clearer imaging.

Educational & Outreach Potential

- Interactive Learning Modules: Apps may include guided tutorials for students or first-time users.
- Community Health Kits: Used in rural outreach programs to enable basic ENT screening with minimal training.
- Gamified Pediatric Mode: Engages children during exams with animated overlays or distraction visuals.

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- History and Evolution of the Otoscope
- Application of Smartphone Otoscope in Telemedicine in Rural Medical Consortium in Eastern China in the COVID-19 Era
- Smartphone Otoscopy Sans Attachment: A Paradigm Shift in Diagnosing Ear Pathologies